

# The Deutsch-Jozsa Algorithm

The First Evidence of Exponential Quantum Speedup

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# The Core Problem

Constant vs. Balanced

Functions

# The Classical Constraint

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$$2^{n-1} + 1$$

QUERIES REQUIRED

## Worst-Case Scenario

For an input string of length  $n$ , a classical computer must query the function at least half plus one times to guarantee a result if the first half are all identical.

This leads to an **exponential growth** in time complexity as  $n$  increases.

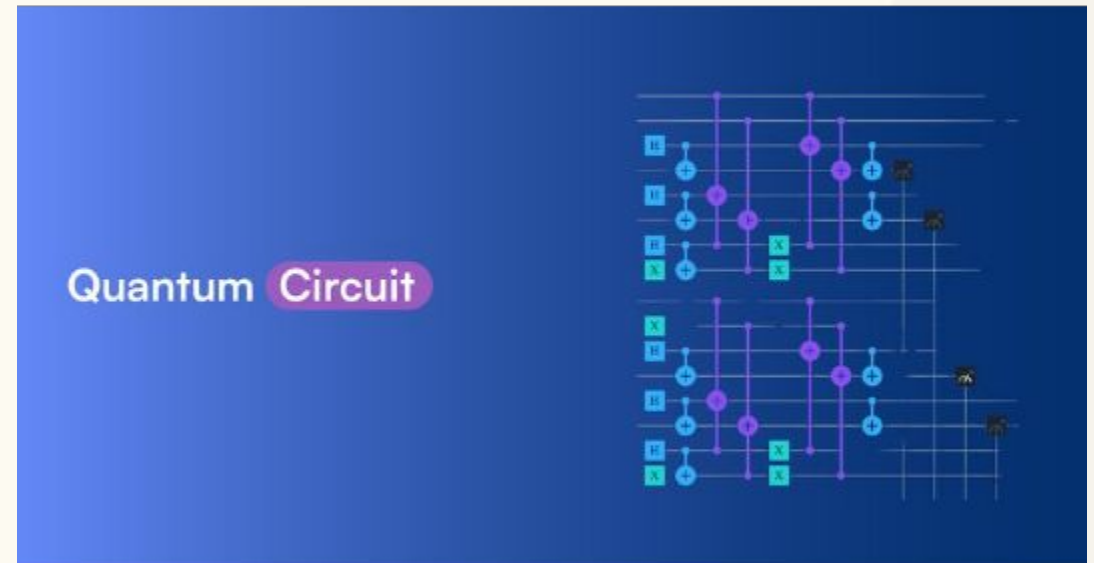
# The Quantum Advantage

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## A Single Query

By leveraging superposition and interference, the Deutsch–Jozsa algorithm determines the function's property in **exactly one** query.

This demonstrates the fundamental power of quantum parallelism, reducing exponential complexity to constant time.



# Defining the Oracle

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## Constant

The function returns the same output for all possible inputs (all 0s or all 1s).

$$f(x) = c$$



## Balanced

The function returns 0 for exactly half of the inputs and 1 for the other half.

$$\sum f(x) = 2^{n-1}$$

# Circuit Architecture

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## Init

Set input   $|0\rangle^n$  and target to  $|1\rangle$ .

## H-Gate

Apply Hadamard to  all qubits for superposition.

## Oracle

Apply the unknown  function  $U_f$ .

## Re-H

Apply Hadamard again to the input  qubits.

## Measure

Observe  the state of the input register.

# Superposition & Initial State

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-  **Massive Parallelism:** Hadamard gates place the input qubits into a uniform superposition of all  $2^n$  possible input strings.
-  **Target Preparation:** The target qubit is placed in the state  $|-\rangle$ , which is crucial for the phase kickback mechanism.
-  **The Global State:** Before the oracle, the system exists as a perfectly balanced product state of all inputs.

# Phase Kickback Mechanism

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## The "Kickback" Effect

When the oracle is applied, the target qubit  $|-\rangle$  remains unchanged, but the phase  $(-1)^{f(x)}$  is "kicked back" onto the input register.

$$U_f|x\rangle|-\rangle = (-1)^{f(x)}|x\rangle|-\rangle$$

## Interference Result

The second set of Hadamard gates causes interference. For **Constant** functions, the phases are constructive at  $|0\rangle$ . For **Balanced**, they cancel out at  $|0\rangle$ .



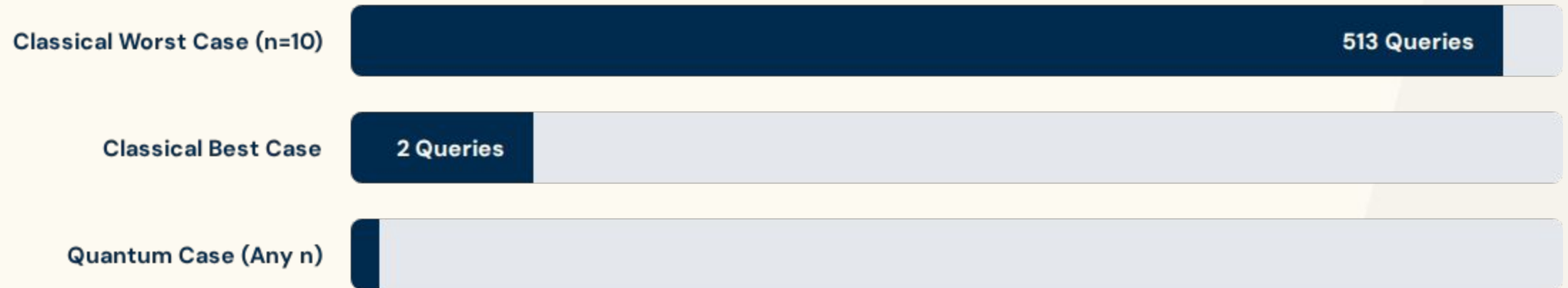
# Mathematical Proof Summary

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| Outcome     | Measured State                                  | Function Property      |
|-------------|-------------------------------------------------|------------------------|
| All Zeroes  | $ 00\dots 0\rangle$                             | <b>Constant</b>        |
| Non-Zero    | Any state other than all 0s                     | <b>Balanced</b>        |
| Probability | $P( 0\rangle^n) =  (1/2^n) \sum (-1)^{f(x)} ^2$ | Unit Prob vs Zero Prob |

# Query Complexity Comparison

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*Quantum complexity is  $O(1)$ , while classical is  $O(2^n-1)$ .*

# Historical Significance

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## Breaking the Barrier

Proposed by David Deutsch and Richard Jozsa in 1992, this was one of the first algorithms to prove that quantum computers could outperform classical Turing machines. While the problem itself is niche, it paved the way for more complex algorithms like Shor's and Grover's.

